Chapter 4. Determinants

One number that decides invertibility and measures volume

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4.1 Determinants of order 2

Motivation. For 2×2 matrices one number, $ad-bc$, decides invertibility and gives the signed area of the parallelogram spanned by the rows. It is not linear in the matrix but is linear in each row separately, and that is the property that generalises.

Definition (Determinant of order 2)

For $A=\left(\begin{smallmatrix} a & b \\ c & d\end{smallmatrix}\right)\in \mathrm{M}_{2\times 2}(\mathbb{F})$, $\det(A)=|A|=ad-bc$.

Example 1

For $A=\left(\begin{smallmatrix} 1 & 2 \\ 3 & 4\end{smallmatrix}\right)$ and $B=\left(\begin{smallmatrix} 3 & 2 \\ 6 & 4\end{smallmatrix}\right)$: $\det A=-2$, $\det B=0$, but $\det(A+B)=\det\left(\begin{smallmatrix} 4 & 4 \\ 9 & 8\end{smallmatrix}\right)=-4\neq \det A+\det B$. So det is not linear.

Theorem 4.1

$\det:\mathrm{M}_{2\times 2}(\mathbb{F})\to\mathbb{F}$ is linear in each row when the other row is fixed: for $u,v,w\in\mathbb{F}^2$, $k\in\mathbb{F}$,

$$\det\begin{pmatrix} u+kv \\ w\end{pmatrix}=\det\begin{pmatrix} u \\ w\end{pmatrix}+k\det\begin{pmatrix} v \\ w\end{pmatrix},\qquad \det\begin{pmatrix} w \\ u+kv\end{pmatrix}=\det\begin{pmatrix} w \\ u\end{pmatrix}+k\det\begin{pmatrix} w \\ v\end{pmatrix}.$$

Proof. With $u=(a_1,a_2)$, $v=(b_1,b_2)$, $w=(c_1,c_2)$: $(a_1+kb_1)c_2-(a_2+kb_2)c_1=(a_1c_2-a_2c_1)+k(b_1c_2-b_2c_1)$. The second identity is analogous. ∎

Theorem 4.2

$A\in\mathrm{M}_{2\times 2}(\mathbb{F})$ is invertible iff $\det A\neq 0$, and then $A^{-1}=\frac{1}{\det A}\left(\begin{smallmatrix} A_{22} & -A_{12} \\ -A_{21} & A_{11}\end{smallmatrix}\right)$.

Proof. If $\det A\neq 0$, multiplying out shows that the displayed matrix M satisfies $AM=MA=I$. Conversely let A be invertible, so $\operatorname{rank}A=2$ and $A_{11}\neq 0$ or $A_{21}\neq 0$. If $A_{11}\neq 0$, add $-A_{21}/A_{11}$ times row 1 to row 2, getting $\left(\begin{smallmatrix} A_{11} & A_{12} \\ 0 & A_{22}-A_{12}A_{21}/A_{11}\end{smallmatrix}\right)$, still of rank 2 (Theorem 3.4). So $A_{22}-A_{12}A_{21}/A_{11}\neq 0$, i.e. $\det A\neq 0$. If $A_{21}\neq 0$, argue the same way with the rows exchanged. ∎

4.1.1 The area of a parallelogram

For an ordered basis $\beta=\{u,v\}$ of $\mathbb{R}^2$, the **orientation** is $O\left(\begin{smallmatrix} u \\ v\end{smallmatrix}\right)=\det\left(\begin{smallmatrix} u \\ v\end{smallmatrix}\right)/\left|\det\left(\begin{smallmatrix} u \\ v\end{smallmatrix}\right)\right|=\pm 1$: $+1$ if β is right-handed (rotating u counterclockwise by less than π reaches the direction of v), -1 if left-handed; set $O=1$ if $\{u,v\}$ is dependent. The **area** of the parallelogram with adjacent sides u,v satisfies

$$\operatorname{area}\begin{pmatrix} u \\ v\end{pmatrix}=O\begin{pmatrix} u \\ v\end{pmatrix}\cdot\det\begin{pmatrix} u \\ v\end{pmatrix}=\left|\det\begin{pmatrix} u \\ v\end{pmatrix}\right|.$$

Reason. $\delta(u,v)=O\cdot\text{area}$ is linear in each row (shearing $v\mapsto v+ku$ preserves area and orientation, and scaling a side scales the signed area), vanishes on equal rows, and $\delta(e_1,e_2)=1$. Such a function equals det (the $n=2$ case of Theorem 4.12).

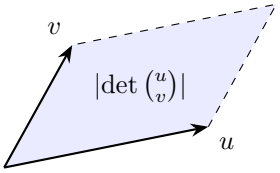


Figure. *The parallelogram spanned by u, v has area $|\det \binom{u}{v}|$; the sign of* det *records orientation.*

Example 2

For $u = (-1, 5)$, $v = (4, -2)$: $\det\begin{pmatrix} -1 & 5 \\ 4 & -2 \end{pmatrix} = 2 - 20 = -18$, so the area is 18 (and $\{u, v\}$ is left-handed).

4.2 Determinants of order n

Motivation. The $n \times n$ determinant is defined recursively by expanding along the first row. From this definition alone one proves linearity in each row and expansion along any row, which yield its behaviour under elementary row operations, the practical way to compute it.

For $A \in \mathrm{M}_{n\times n}(\mathbb{F})$, $\tilde{A}_{ij}$ is the $(n-1) \times (n-1)$ matrix obtained by deleting row i and column j.

Definition (Determinant)

For $n = 1$, $\det(A_{11}) = A_{11}$. For $n \geq 2$,

$$\det(A) = \sum_{j=1}^{n} (-1)^{1+j} A_{1j} \det(\tilde{A}_{1j}).$$

$c_{ij} = (-1)^{i+j} \det(\tilde{A}_{ij})$ is the **cofactor** of A_{ij}, and the formula is the **cofactor expansion along the first row**. For $n = 2$ it gives $A_{11}A_{22} - A_{12}A_{21}$.

Example 1

$A = \begin{pmatrix} 1 & 3 & -3 \\ -3 & -5 & 2 \\ -4 & 4 & -6 \end{pmatrix}$: $\det A = 1 \cdot (30 - 8) - 3 \cdot (18 + 8) + (-3)(-12 - 20) = 22 - 78 + 96 = 40$.

Example 2

$B = \begin{pmatrix} 0 & 1 & 3 \\ -2 & -3 & -5 \\ 4 & -4 & 4 \end{pmatrix}$: $\det B = 0 - 1 \cdot (-8 + 20) + 3 \cdot (8 + 12) = 48$.

Example 3

$C = \begin{pmatrix} 2 & 0 & 0 & 1 \\ 0 & 1 & 3 & -3 \\ -2 & -3 & -5 & 2 \\ 4 & -4 & 4 & -6 \end{pmatrix}$: only C_{11} and C_{14} contribute, and $\tilde{C}_{11} = A$, $\tilde{C}_{14} = B$ above, so $\det C = 2 \cdot 40 + (-1)^5 \cdot 1 \cdot 48 = 32$.

Example 4

$\det I_n = 1$, by induction: the expansion along the first row has only the term $1 \cdot \det I_{n-1}$.

Theorem 4.3

The determinant is linear in each row when the other rows are fixed: if rows $a_1, \ldots, a_n$ have

$a_r = u + kv$, then

$$\det(a_1, \dots, u + kv, \dots, a_n) = \det(a_1, \dots, u, \dots, a_n) + k \det(a_1, \dots, v, \dots, a_n).$$

Proof. Induction on n; $n = 1$ is clear. Let A, B, C be the matrices with row r equal to $u + kv$, u, v and the other rows equal.

Step 1 ($r = 1$). $\tilde{A}_{1j} = \tilde{B}_{1j} = \tilde{C}_{1j}$, so $\det A = \sum_j (-1)^{1+j}(u_j + kv_j) \det \tilde{A}_{1j} = \det B + k \det C$.

Step 2 ($r > 1$). $A_{1j} = B_{1j} = C_{1j}$, and $\tilde{A}_{1j}$, $\tilde{B}_{1j}$, $\tilde{C}_{1j}$ agree except in the row coming from row r, which is $u + kv$, u, v with entry j deleted. By induction $\det \tilde{A}_{1j} = \det \tilde{B}_{1j} + k \det \tilde{C}_{1j}$; multiply by $(-1)^{1+j} A_{1j}$ and sum. ∎

Corollary

A matrix with a zero row has determinant 0.

Proof. If row r is $0 = 0 + 1 \cdot 0$, Theorem 4.3 gives $\det A = \det A + \det A$. ∎

Lemma

If $n \geq 2$ and row i of $B \in \mathrm{M}_{n \times n}(\mathbb{F})$ is e_k, then $\det B = (-1)^{i+k} \det(\tilde{B}_{ik})$.

Proof. Induction on n. For $n = 2$ check directly. Let $n \geq 3$. If $i = 1$, the definition gives the claim. Let $i > 1$. For each $j \neq k$, let C_{ij} be $\tilde{B}_{1j}$ with row $i - 1$ (coming from row i of B) and its column through k removed. Row $i - 1$ of $\tilde{B}_{1j}$ is 0 if $j = k$, e_{k-1} if $j < k$, and e_k if $j > k$ (indices in $\mathbb{F}^{n-1}$). By induction,

$$\det \tilde{B}_{1j} = \begin{cases} 0 & j = k \\ (-1)^{(i-1)+(k-1)} \det C_{ij} & j < k \\ (-1)^{(i-1)+k} \det C_{ij} & j > k. \end{cases}$$

Hence $\det B = \sum_{j<k} (-1)^{1+j} B_{1j} (-1)^{i+k} \det C_{ij} + \sum_{j>k} (-1)^{1+j} B_{1j} (-1)^{i+k+1} \det C_{ij}$. On the other hand, expand $\det \tilde{B}_{ik}$ along its first row (row 1 of B without entry k). The entry B_{1j} sits in column j if $j < k$ and column $j - 1$ if $j > k$, and its minor is C_{ij}. So

$$\det \tilde{B}_{ik} = \sum_{j<k} (-1)^{1+j} B_{1j} \det C_{ij} + \sum_{j>k} (-1)^{j} B_{1j} \det C_{ij}.$$

Multiplying by $(-1)^{i+k}$ gives the expression for $\det B$. ∎

Theorem 4.4

The determinant can be expanded along any row: for each i, $\det A = \sum_{j=1}^{n} (-1)^{i+j} A_{ij} \det(\tilde{A}_{ij})$.

Proof. For $i = 1$ this is the definition; let $i > 1$. Row i of A is $\sum_j A_{ij} e_j$. Let B_j be A with row i replaced by e_j. By Theorem 4.3 and the Lemma, $\det A = \sum_j A_{ij} \det B_j = \sum_j A_{ij} (-1)^{i+j} \det(\tilde{B}_j)_{ij} = \sum_j (-1)^{i+j} A_{ij} \det \tilde{A}_{ij}$, since deleting row i removes the only difference between B_j and A. ∎

Corollary

A matrix with two identical rows has determinant 0.

Proof. Induction on n; for $n = 2$ it is clear. For $n \geq 3$, if rows $r \neq s$ coincide, expand along a row $i \notin \{r, s\}$: each $\tilde{A}_{ij}$ has two identical rows, so every term vanishes by induction. ∎

Theorem 4.5

Interchanging two rows multiplies the determinant by -1.

Proof. Let rows $r < s$ of A be u, v and B have them interchanged. By Theorem 4.3 and the Corollary, $0 = \det(\dots, u+v, \dots, u+v, \dots) = \det(\dots, u, \dots, u, \dots) + \det(\dots, u, \dots, v, \dots) + \det(\dots, v, \dots, u, \dots) + \det(\dots, v, \dots, v, \dots) = 0 + \det A + \det B + 0$. ■

Theorem 4.6

Adding a multiple of one row to another leaves the determinant unchanged.

Proof. Let B be A with k·row r added to row s, and C be A with row s replaced by row r. By Theorem 4.3, $\det B = \det A + k \det C = \det A$, since $\det C = 0$ by the Corollary to Theorem 4.4. ■

Corollary

If $\operatorname{rank} A < n$, then $\det A = 0$.

Proof. The rows are dependent (Corollary 2 of Theorem 3.6), so some row $a_r = \sum_{i \neq r} c_i a_i$. Subtracting c_i·row i from row r for each $i \neq r$ produces a zero row without changing det (Theorem 4.6); apply the Corollary to Theorem 4.3. ■

Summary for row operations: type 1 multiplies det by -1; type 2 (row times k) multiplies it by k; type 3 leaves it unchanged. The determinant of an upper triangular matrix is the product of its diagonal entries (expand along the last row repeatedly), so reducing to triangular form computes det.

Example 5

For B of Example 2: swap rows $1, 2$ (sign change), clear column 1, and reach the triangular matrix $\begin{pmatrix} -2 & -3 & -5 \\ 0 & 1 & 3 \\ 0 & 0 & 24 \end{pmatrix}$. So $\det B = -(-2)(1)(24) = 48$.

Example 6

For $\begin{pmatrix} 2 & 3 & 1 & 1 \\ -2 & -2 & 2 & -1 \\ 4 & 7 & 2 & 3 \\ -6 & -9 & 5 & 1 \end{pmatrix}$, type 3 operations give an upper triangular matrix with diagonal product -40, the determinant.

4.3 Properties of determinants

Motivation. The determinant is multiplicative, detects invertibility, ignores transposition, solves square systems (Cramer's rule), and measures n-dimensional volume.

From the row-operation rules: $\det E = -1$, k, 1 for elementary E of types $1, 2, 3$, and $\det(EA) = \det E \cdot \det A$ for elementary E.

Theorem 4.7

$\det(AB) = \det(A) \det(B)$ for $A, B \in \mathrm{M}_{n \times n}(\mathbb{F})$.

Proof.

Step 1 ($\operatorname{rank} A < n$). $\operatorname{rank}(AB) \le \operatorname{rank} A < n$ (Theorem 3.7), so both sides are 0 (Corollary to Theorem 4.6).

Step 2 ($\operatorname{rank} A = n$). $A = E_m \cdots E_1$ is a product of elementary matrices (Corollary 3 of Theorem 3.6). Applying $\det(EM) = \det E \det M$ repeatedly, $\det(AB) = \det E_m \cdots \det E_1 \det B$ and $\det A = \det E_m \cdots \det E_1$. ■

Corollary

A is invertible iff $\det A \neq 0$, and then $\det(A^{-1}) = \frac{1}{\det A}$.

Proof. If A is not invertible, $\operatorname{rank} A < n$ and $\det A = 0$. If A is invertible, $1 = \det I = \det(AA^{-1}) = \det A \det A^{-1}$. ■

Theorem 4.8

$\det(A^t) = \det(A)$.

Proof. If A is not invertible, neither is A^t ($\operatorname{rank} A^t = \operatorname{rank} A$), and both determinants are 0. Otherwise $A = E_m \cdots E_1$, so $A^t = E_1^t \cdots E_m^t$. Each E_i^t is elementary of the same type with $\det E_i^t = \det E_i$ (types 1, 2 are symmetric; a type 3 transpose is again type 3). By Theorem 4.7, $\det A^t = \prod \det E_i^t = \prod \det E_i = \det A$. ■

Consequently all row statements hold for columns: expansion along any column, and the effect of column operations.

Theorem 4.9 (Cramer's rule)

If $Ax = b$ is an $n \times n$ system with $\det A \neq 0$, it has a unique solution, and

$$x_k = \frac{\det M_k}{\det A} \qquad (1 \le k \le n),$$

where M_k is A with column k replaced by b.

Proof. Uniqueness: A is invertible (Theorem 3.10). Let x be the solution and X_k be I_n with column k replaced by x. Then $AX_k = M_k$ (column $j \neq k$ of AX_k is $Ae_j = a_j$, column k is $Ax = b$). Expanding along row k, $\det X_k = x_k$. So $\det M_k = \det A \cdot x_k$. ■

Example 1

$x_1 + 2x_2 + 3x_3 = 2$, $x_1 + x_3 = 3$, $x_1 + x_2 - x_3 = 1$: $\det A = 6$, $\det M_1 = 15$, $\det M_2 = -6$, $\det M_3 = 3$, so $(x_1, x_2, x_3) = (\frac{5}{2}, -1, \frac{1}{2})$.

Volume. For rows $a_1, \ldots, a_n \in \mathbb{R}^n$ of A, $|\det A|$ is the n-dimensional volume of the parallelepiped with adjacent sides $a_1, \ldots, a_n$ (this follows as in Section 4.1.1, the multilinear-alternating argument of Section 4.5).

Example 2

$a_1 = (1, -2, 1)$, $a_2 = (1, 0, -1)$, $a_3 = (1, 1, 1)$ span a parallelepiped of volume $\left|\det \begin{pmatrix} 1 & -2 & 1 \\ 1 & 0 & -1 \\ 1 & 1 & 1 \end{pmatrix}\right| = 6$. (These vectors are mutually orthogonal, of lengths $\sqrt{6}, \sqrt{2}, \sqrt{3}$, and $\sqrt{6}\sqrt{2}\sqrt{3} = 6$.)

For an ordered basis β of $\mathbb{R}^n$ with vectors as rows of A, β is **right-handed** if $\det A > 0$ and **left-handed** if $\det A < 0$. A change of coordinates with $\det Q > 0$ preserves handedness.

4.4 Summary: important facts about determinants

Motivation. The facts needed later, collected in one place.

For $A \in \mathrm{M}_{n\times n}(\mathbb{F})$:

1. $\det A = \sum_j (-1)^{i+j} A_{ij} \det \tilde{A}_{ij}$ for any i, and $= \sum_i (-1)^{i+j} A_{ij} \det \tilde{A}_{ij}$ for any j (Theorems 4.4, 4.8).

2. Interchanging two rows (columns) changes the sign; multiplying a row (column) by k multiplies det by k; adding a multiple of a row (column) to another leaves it unchanged.

3. det of an upper or lower triangular matrix is the product of the diagonal entries; $\det I = 1$.

4. Two identical rows (columns), or a zero row, give det $= 0$.

5. $\det(AB) = \det A \det B$; $\det A^t = \det A$.

6. A is invertible iff $\det A \neq 0$, and then $\det A^{-1} = (\det A)^{-1}$.

7. Similar matrices have equal determinants: $\det(Q^{-1}AQ) = \det Q^{-1} \det A \det Q = \det A$. Hence the **determinant of a linear operator** $\det \mathsf{T} = \det[\mathsf{T}]_\beta$ is independent of β.

8. For $A = \left(\begin{smallmatrix} B & C \\ O & D \end{smallmatrix}\right)$ with square B, D, $\det A = \det B \det D$.

Proof of 3 and 8 [Supplement].

3. For upper triangular A expand along the first column: $\det A = A_{11} \det \tilde{A}_{11}$, and $\tilde{A}_{11}$ is upper triangular; induct. Lower triangular: transpose.

8. Induction on the size k of B, expanding along the first column. For $k = 1$, $\det A = B_{11} \det D$. For $k > 1$, only rows $i \leq k$ contribute, and $\tilde{A}_{i1} = \left(\begin{smallmatrix} \tilde{B}_{i1} & C' \\ O & D \end{smallmatrix}\right)$, so $\det A = \sum_{i\leq k} (-1)^{i+1} B_{i1} \det \tilde{B}_{i1} \det D = \det B \det D$. ∎

4.5 A characterization of the determinant *

Motivation. Linearity in each row, vanishing on matrices with two equal adjacent rows, and the value 1 at I determine the determinant uniquely. This explains why det measures signed volume and gives a second proof of multiplicativity.

Definition (n-linear; alternating)

$\delta : \mathrm{M}_{n\times n}(\mathbb{F}) \to \mathbb{F}$ is **n-linear** if it is linear in each row when the others are fixed. It is **alternating** if $\delta(A) = 0$ whenever two adjacent rows of A are identical.

Example 1

$\delta(A) = 0$ is n-linear.

Example 2

$\delta_j(A) = A_{1j}A_{2j}\cdots A_{nj}$ (product of column j) is n-linear: it is linear in A_{rj}, i.e. in row r.

Example 3

$\delta(A) = A_{11}A_{22}\cdots A_{nn}$ is n-linear.

Example 4

tr is not n-linear for $n \geq 2$: $\mathrm{tr}(2I) = 2n \neq 2\mathrm{tr}(I)$, whereas an n-linear function would scale $2I$ (each of n rows doubled) by 2^n.

det is n-linear (Theorem 4.3) and alternating (Corollary to Theorem 4.4).

Theorem 4.10

Let δ be alternating and n-linear.
(a) Interchanging two rows of A changes $\delta(A)$ to $-\delta(A)$.
(b) If A has two identical rows, $\delta(A) = 0$.

Proof.

(a) For adjacent rows $r, r+1$ with entries u, v, the computation of Theorem 4.5 works because only adjacent equal rows are used: $0 = \delta(\dots, u+v, u+v, \dots) = \delta(A) + \delta(B)$. A swap of rows $r < s$ is a composition of $2(s-r)-1$ adjacent swaps (move row r down to position s, then row s up to position r), an odd number. So δ changes sign.

(b) Swap rows to make the identical rows adjacent; by (a) this changes δ only by a sign, and the result is 0. ■

Corollary 1

If B is A with a multiple of a row added to another row, then $\delta(B) = \delta(A)$.

Proof. As for Theorem 4.6, using Theorem 4.10(b). ■

Corollary 2

If rank $M < n$, then $\delta(M) = 0$.

Proof. As for the Corollary to Theorem 4.6: row operations of type 3 (Corollary 1) produce a zero row, and δ of a matrix with a zero row is 0 by linearity. ■

Corollary 3

For elementary E_1, E_2, E_3 of types 1, 2, 3, with E_2 multiplying a row of I by k: $\delta(E_1) = -\delta(I)$, $\delta(E_2) = k\delta(I)$, $\delta(E_3) = \delta(I)$.

Proof. Theorem 4.10(a), linearity in the scaled row, and Corollary 1, applied to I. ■

Theorem 4.11

If δ is alternating, n-linear and $\delta(I) = 1$, then $\delta(AB) = \delta(A)\delta(B)$.

Proof [Supplement]. By Theorem 4.10, linearity and Corollary 1, $\delta(EB) = \delta(E)\delta(B)$ for elementary E (with $\delta(E) = -1, k, 1$ by Corollary 3). Now repeat the proof of Theorem 4.7, using Corollary 2 when rank $A < n$. ■

Theorem 4.12

If δ is alternating, n-linear and $\delta(I) = 1$, then $\delta(A) = \det(A)$ for all A.

Proof. If $\operatorname{rank} A < n$, both vanish (Corollary 2 and the Corollary to Theorem 4.6). Otherwise $A = E_m \cdots E_1$, and by Theorem 4.11 and Corollary 3, $\delta(A) = \prod \delta(E_i) = \prod \det(E_i) = \det A$. ■

So det is the *unique* alternating n-linear function with value 1 at I.